

Problem #1 – Combined Loading

1. Draw plane square/circle
2. Define axis. Find Normal, Forces, Moments, Torques
4. Find $\sigma_{Ax}, \sigma_{Ay}, \tau_A, \sigma_{Bx}, \sigma_{By}, \tau_B$ using the info below:

Causes of σ : $\frac{Pr}{t}, \frac{Pr}{2t}, \frac{N}{A}, \frac{Mc}{I}$

Causes of τ : $\frac{VQ}{It}$ (Check for zero Q bc of $\perp V$), $\frac{Tc}{J}$

$$I_x = \frac{1}{12}bh^3, I_y = \frac{1}{12}b^3h, I_o = \frac{\pi}{4}r^4$$

$$Q = \bar{y}A, Q_{\bullet} = \frac{2}{3}r^3$$

$$J_o = \frac{\pi}{2}r^4$$

Problem #1 + #3 - Mohr Circle in Combined Loading

5. Draw Axes:

$$6. P_1: [\sigma_{xA}, \tau_A], P_2: [\sigma_{yA}, -\tau_A]$$

7. Draw line from P1 to P2 for Diameter.

$$8. \text{Label } R = \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_A^2}$$

$$9. \text{Label } O (\text{center}) = \left(\sigma_{\text{avg}} = \frac{\sigma_x + \sigma_y}{2}, 0\right)$$

10. Principal Stresses:

$$\text{Max PS} = \sigma_1 = \sigma_{\text{avg}} + R$$

$$\text{Min PS} = \sigma_2 = \sigma_{\text{avg}} - R$$

Problem #2 – Pressure Vessel (Spiral Weld)

1. Check for thin wall condition: $\frac{r}{t} > 10$
2. Draw Initial Stress Element:

3. Calculate the following:

$$\sigma_{\text{hoop}} = \sigma_y = \frac{Pr}{t} = \underline{\hspace{2cm}}$$

$$\sigma_{\text{longitudinal}} = \sigma_x = \frac{Pr}{2t} = \underline{\hspace{2cm}}$$

$$\tau_{xy} = 0 \text{ (usually)}$$

4. Draw the Stress State Element:

5. Calculate the following:

$$\sigma_{x'} = \frac{\sigma_x + \sigma_y}{2} + \frac{\sigma_x - \sigma_y}{2} \cos(2\theta) + \tau_{xy} \sin(2\theta)$$

$$\sigma_{y'} = \frac{\sigma_x + \sigma_y}{2} - \frac{\sigma_x - \sigma_y}{2} \cos(2\theta) - \tau_{xy} \sin(2\theta)$$

$$\tau_{x'y'} = -\left(\frac{\sigma_x - \sigma_y}{2}\right) \sin(2\theta) + \tau_{xy} \cos(2\theta)$$

Problem #4 – Mohr's Circle, Principal and Max Stress

1. Find $\sigma_x, \sigma_y, \tau_{xy}$ from Initial Stress Element (T+, C-):

2. Find Points 1 and 2 (CCW $\curvearrowright$ (+), CW $\curvearrowleft$ (-)):

$$\text{Point 1: } (\sigma_x, \tau_{xy}/\text{direction}) = (\underline{\hspace{1cm}}, \underline{\hspace{1cm}})$$

$$\text{Point 2: } (\sigma_y, \tau_{xy}/\text{direction}) = (\underline{\hspace{1cm}}, \underline{\hspace{1cm}})$$

3. Draw x and y axis for Mohr's Circle

4. Plot Points 1 and 2. Connect together for diameter.

5. Calculate then label the Circle Center, O:

$$\sigma_{\text{avg}} = \frac{\sigma_x + \sigma_y}{2} = \underline{\hspace{2cm}}$$

$$O = (\sigma_{\text{avg}}, 0) = \left(\frac{\sigma_x + \sigma_y}{2}, 0\right) = (\underline{\hspace{1cm}}, 0)$$

6. Calculate then label the radius, R:

$$R = \sqrt{\left(\sigma_{\text{avg}}\right)^2 + (\tau_{xy})^2} = \tau_{\text{max}} / \text{direction} = \underline{\hspace{2cm}}$$

7. Draw the circle

8. Label Vertical Diameter:

$$(\sigma_{\text{avg}}, \tau_{\text{max-cw}}) = \text{max shear stress}$$

9. Identify σ_1 and σ_2 with:

$$\text{Max Principal Stress} = \sigma_1 = \sigma_{\text{avg}} + R = \underline{\hspace{2cm}}$$

$$\text{Min Principal Stress} = \sigma_2 = \sigma_{\text{avg}} - R = \underline{\hspace{2cm}}$$

10. Label a and b as legs for Point O to P1 right triangle:

$$a = O_y - P1_y = \underline{\hspace{2cm}}$$

$$b = O_x - P1_x = \underline{\hspace{2cm}}$$

11. Find θ_p and θ_s using the following:

$$\tan(2\theta_p) = \frac{b}{a} \text{ and } \tan(2\theta_s) = \frac{a}{b}$$

12. Draw Principal Stress Element

13. Draw Maximum Shear Stress Element

Problem #4 – Mohr's Circle, CONT'D:

14. Calculate $\tau_{\text{absolute-max}}$ using the following:

If σ_1 and σ_2 have **opposite** signs: $\tau_{\text{abs-max}} = \frac{\sigma_1 - \sigma_2}{2}$

If σ_1 and σ_2 have **similar** signs: $\tau_{\text{abs-max}} = \frac{\sigma_1}{2}$

Cool Facts

$$\text{Pa} = \frac{N}{m^2}, \text{kPa} = \frac{N}{mm^2}$$

$$\text{psi} = \frac{lb}{in^2}$$

